

# LD11 Glossary

Detailed glossary of key Day 11 hyperparameter tuning terms.

| Field         | Value                                   |
|---------------|-----------------------------------------|
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**Submission note**

This is an individual PDF document. The documentation package intentionally does not include a combined/merged PDF.

# 1. Glossary

| Term                  | Detailed simple meaning                                                                                                      |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------|
| Hyperparameter        | A setting that controls model behaviour before or during training. Examples: n_neighbors, alpha, C, max_depth, n_estimators. |
| Parameter             | A value learned from the data during training. Examples: coefficients, weights, split thresholds.                            |
| Validation score      | The score used to compare hyperparameter settings during tuning.                                                             |
| Test score            | The final score measured on data not used for tuning decisions.                                                              |
| GridSearchCV          | A cross-validated search method that tests all parameter combinations from a grid.                                           |
| RandomizedSearchCV    | A cross-validated search method that samples a fixed number of parameter combinations.                                       |
| KNN                   | K-Nearest Neighbours, a distance-based algorithm that predicts using nearby training points.                                 |
| StandardScaler        | A preprocessing method that centers and scales numeric features. Critical for distance-based models.                         |
| Logistic Regression   | A linear classification model that estimates class probabilities.                                                            |
| C                     | Inverse regularisation strength in Logistic Regression. Smaller C means stronger regularisation.                             |
| Ridge Regression      | Linear regression with L2 regularisation to shrink coefficients.                                                             |
| Alpha                 | Regularisation strength in Ridge Regression.                                                                                 |
| CASH                  | Combined Algorithm Selection and Hyperparameter Optimization.                                                                |
| Objective function    | The function an optimiser tries to minimise or maximise during tuning.                                                       |
| Hyperopt              | A library for hyperparameter optimisation with search spaces and objective functions.                                        |
| Optuna                | A modern tuning framework based on studies and trials.                                                                       |
| Trial                 | One tested hyperparameter configuration.                                                                                     |
| Bayesian optimisation | A method that uses previous trials and uncertainty to decide what to test next.                                              |
| Surrogate model       | A model that approximates the relationship between hyperparameters and score.                                                |
| Exploration           | Trying uncertain or less-tested areas of the search space.                                                                   |
| Exploitation          | Trying near already strong areas of the search space.                                                                        |
| Acquisition function  | A rule that chooses the next Bayesian optimisation trial.                                                                    |
| Expected Improvement  | An acquisition idea that measures expected gain over the current best result.                                                |
| NAS                   | Neural Architecture Search, the process of searching neural-network structures.                                              |
| Successive Halving    | A multi-fidelity method that starts many candidates and removes weak ones early.                                             |
| Hyperband             | A method that repeats Successive Halving with different resource budgets.                                                    |
| AutoML                | Automation of parts of the ML workflow such as preprocessing, model selection, tuning, and ensembling.                       |